Resistance Conjecture Quoted in arXiv:2509.04155 Answered by arXiv:2602.05477

Run fa_banach_001, agent_lane_14

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Status

This is a lightweight literature-status packet:

`literature_already_answered`

for the resistance conjecture as quoted in Riku Anttila, "An Approach to Sub-Gaussian Heat Kernel Estimates via Analysis on Metric Spaces", arXiv:2509.04155.

Source Question

In Section 7, arXiv:2509.04155 discusses the resistance conjecture of Grigor'yan, Hu, and Lau and states it as Conjecture 4.15. In the notation of that paper:

> The strongly local regular Dirichlet form $(\mathcal{E}, \mathcal{F})$ on $L^2(X, \mu)$ satisfies the heat kernel estimates if and only if $(\mathcal{E}, \mathcal{F})$ satisfies both the Poincare inequality and the upper capacity estimate.

The surrounding text explains that the conjecture asks whether the cutoff energy condition in Theorem 6.4 can be replaced by the upper capacity estimate.

Supporting Answer

Sylvester Eriksson-Bique, "On the Resistance Conjecture", arXiv:2602.05477, explicitly states in the abstract that it gives an affirmative answer to the resistance conjecture. Its Theorem 1.1 proves the equivalence, for metric measure Dirichlet spaces, of:

1. parabolic Harnack inequality PHI(β);
2. heat-kernel estimates HKE(β) together with volume doubling;
3. volume doubling, elliptic Harnack inequality, and a two-sided capacity bound;
4. volume doubling, Poincare inequality PI(β), and the capacity upper bound $\mathrm{Cap}_2(\beta)$.

The paper further explains that its key step is proving that volume doubling, Poincare, and upper capacity imply the cutoff Sobolev inequality.

Thus the later paper supplies the advertised replacement of cutoff Sobolev by upper capacity bounds, in the standard volume-doubling formulation. The supporting paper also cites arXiv:2509.04155 and describes Anttila's work as a closely related earlier approach with partial results.

Scope

This packet records a literature answer only; it is not a new proof. It does not settle the sharper question isolated in the existing attempt for arXiv:2602.05477, namely whether upper capacity plus a separated-ball lower capacity condition and volume doubling imply the Poincare inequality itself.

Search Evidence

Cheap run indexes showed no exact prior packet for arXiv:2509.04155. They did show an arXiv:2602.05477 attempt noting that the later paper solves the headline resistance conjecture and leaves a sharper capacity-to-Poincare question open. Local source inspection matched the Conjecture 4.15 discussion in arXiv:2509.04155 with the explicit affirmative-answer language and Theorem 1.1 of arXiv:2602.05477.

References

[1] Riku Anttila, *An Approach to Sub-Gaussian Heat Kernel Estimates via Analysis on Metric Spaces*, arXiv:2509.04155.

[2] Sylvester Eriksson-Bique, *On the Resistance Conjecture*, arXiv:2602.05477.